241801053-J B Dhejasvi Branch and Bound

```
from itertools import product
```

1. Maximize Z = 5x + 4y Subject to: 2x + y ≤ 8 x + 2y ≤ 8 x, y ≥ 0 and integers.

```
best = None

for x in range(0, 100):
    for y in range(0, 100):
        if 2*x + y <= 8 and x + 2*y <= 8:
            Z = 5*x + 4*y
            if best is None or Z > best[0]:
                best = (Z, x, y)

print("Optimal (Z, x, y):", best)
```

```
Optimal (Z, x, y): (23, 3, 2)
```

2. Minimize Z = 6x + 5y Subject to: 3x + 2y ≥ 12 x + y ≥ 5 x, y ≥ 0 and integers.

```
best = None  # (Z, x, y)

for x in range(0, 100):
    for y in range(0, 100):
        if 3*x + 2*y >= 12 and x + y >= 5:
            Z = 6*x + 5*y
            if best is None or Z < best[0]:
                best = (Z, x, y)

print("Optimal (Z, x, y):", best)
```

```
Optimal (Z, x, y): (27, 2, 3)
```

3. Maximize Z = 4x1 + 3x2 + 2x3 Subject to: x1 + x2 + x3 ≤ 5 2x1 + x2 ≤ 6 x1, x2, x3 ≥ 0 and integers.

```
best = None  # (Z, x1, x2, x3)

for x1 in range(0, 100):
    for x2 in range(0, 100):
        for x3 in range(0, 100):
            if (x1 + x2 + x3 <= 5) and (2*x1 + x2 <= 6):
                Z = 4*x1 + 3*x2 + 2*x3
                if best is None or Z > best[0]:
                    best = (Z, x1, x2, x3)

print("Optimal (Z, x1, x2, x3):", best)
```

```
Optimal (Z, x1, x2, x3): (16, 1, 4, 0)
```

4. Maximize Z = 10x1 + 6x2 + 4x3 Subject to: 5x1 + 4x2 + 2x3 ≤ 8 x1, x2, x3 ∈ {0,1}.

```
from itertools import product

best = None  # (Z, x1, x2, x3)

for x1, x2, x3 in product([0, 1], repeat=3):
    if 5*x1 + 4*x2 + 2*x3 <= 8:
        Z = 10*x1 + 6*x2 + 4*x3
        if best is None or Z > best[0]:
            best = (Z, x1, x2, x3)

print("Optimal (Z, x1, x2, x3):", best)
```

```
Optimal (Z, x1, x2, x3): (14, 1, 0, 1)
```

5. Maximize Z = 8x + 5y Subject to: 4x + 3y ≤ 24 x ∈ Z, y ≥ 0.

```
best = None  # (Z, x, y)

for x in range(-50, 51):   # safe range
    rhs = 24 - 4*x
    if rhs >= 0:
        y = rhs / 3.0      # max y possible
        Z = 8*x + 5*y
        if best is None or Z > best[0]:
            best = (Z, x, y)

print("Optimal (Z, x, y):", best)
```

```
Optimal (Z, x, y): (48.0, 6, 0.0)
```

6. A factory produces Product A and Product B. Product Profit per unit (₹) Machine hours required A 40 2 B 30 1 The factory has 7 machine hours available per day. Products must be produced in whole units. (a) Formulate the problem as an Integer Linear Programming model. (b) Solve the model using the Branch and Bound method. (c) Determine the optimal production plan and maximum profit.

```
best = None  # (profit, xA, xB)

for xA in range(0, 100):
    for xB in range(0, 100):
        if 2*xA + xB <= 7:          # machine hours constraint
            profit = 40*xA + 30*xB  # objective
            if best is None or profit > best[0]:
                best = (profit, xA, xB)

print("Best (profit, xA, xB):", best)
```

```
Best (profit, xA, xB): (210, 0, 7)
```

7. A company is considering three independent investment projects.

Project	Investment required (₹ lakh)	Expected profit (₹ lakh)
1	5	9
2	4	6
3	3	5

The total available capital is ₹10 lakhs. Each project can be either accepted or rejected.

(a) Formulate the problem as a Binary Integer Programming model.
(b) Solve the problem using the Branch and Bound method.
(c) Identify the selected projects and the maximum total profit.

Model Answers

```
investment = [5, 4, 3]
profit     = [9, 6, 5]
budget = 10

best = None

for x1 in [0, 1]:
    for x2 in [0, 1]:
        for x3 in [0, 1]:
            total_inv = x1*investment[0] + x2*investment[1] + x3*investment[2]
            if total_inv <= budget:
                total_profit = x1*profit[0] + x2*profit[1] + x3*profit[2]
                if best is None or total_profit > best[0]:
                    best = (total_profit, x1, x2, x3, total_inv)

print("Best (profit, x1, x2, x3, investment):", best)

total_profit, x1, x2, x3, total_inv = best
selected = [i+1 for i, x in enumerate([x1, x2, x3]) if x == 1]
print("Selected projects:", selected)
print("Total investment:", total_inv, "lakhs")
print("Maximum profit:", total_profit, "lakhs")
```

```
Best (profit, x1, x2, x3, investment): (15, 1, 1, 0, 9)
Selected projects: [1, 2]
Total investment: 9 lakhs
Maximum profit: 15 lakhs
```